



## LNP\* THERMOCOMP\* Compound DX14354X

### Asia Pacific: COMMERCIAL

LNP™ Thermocomp™ compound DX14354X is an improved flow, colorable compound based on PC copolymer resin developed for applications that require Laser Direct Structuring (LDS) for antenna, or electronic circuit manufacturing. Thermocomp compound DX14354X helps customers to improve productivity with stable plating and RF performance, excellent impact strength and surface finish.

| TYPICAL PROPERTIES <sup>1</sup>              | TYPICAL VALUE | UNIT              | STANDARD    |
|----------------------------------------------|---------------|-------------------|-------------|
| <b>MECHANICAL</b>                            |               |                   |             |
| Tensile Stress, yld, Type I, 50 mm/min       | 54            | MPa               | ASTM D 638  |
| Tensile Stress, brk, Type I, 50 mm/min       | 44            | MPa               | ASTM D 638  |
| Tensile Strain, yld, Type I, 50 mm/min       | 5             | %                 | ASTM D 638  |
| Tensile Strain, brk, Type I, 50 mm/min       | 70            | %                 | ASTM D 638  |
| Tensile Modulus, 50 mm/min                   | 2540          | MPa               | ASTM D 638  |
| Flexural Stress, yld, 1.3 mm/min, 50 mm span | 85            | MPa               | ASTM D 790  |
| Flexural Stress, brk, 1.3 mm/min, 50 mm span | 83            | MPa               | ASTM D 790  |
| Flexural Modulus, 1.3 mm/min, 50 mm span     | 2530          | MPa               | ASTM D 790  |
| Tensile Stress, yield, 50 mm/min             | 54            | MPa               | ISO 527     |
| Tensile Stress, break, 50 mm/min             | 47            | MPa               | ISO 527     |
| Tensile Strain, yield, 50 mm/min             | 5             | %                 | ISO 527     |
| Tensile Strain, break, 50 mm/min             | 80            | %                 | ISO 527     |
| Tensile Modulus, 1 mm/min                    | 2460          | MPa               | ISO 527     |
| Flexural Stress, yield, 2 mm/min             | 86            | MPa               | ISO 178     |
| Flexural Modulus, 2 mm/min                   | 2560          | MPa               | ISO 178     |
| <b>IMPACT</b>                                |               |                   |             |
| Izod Impact, notched, 23°C                   | 800           | J/m               | ASTM D 256  |
| Izod Impact, notched 80*10*3 +23°C           | 60            | kJ/m <sup>2</sup> | ISO 180/1A  |
| <b>THERMAL</b>                               |               |                   |             |
| Vicat Softening Temp, Rate A/50              | 128           | °C                | ASTM D 1525 |
| Vicat Softening Temp, Rate B/50              | 128           | °C                | ASTM D 1525 |
| HDT, 1.82 MPa, 3.2mm, unannealed             | 111           | °C                | ASTM D 648  |
| CTE, -40°C to 40°C, flow                     | 5.7E-05       | 1/°C              | ASTM E 831  |
| CTE, -40°C to 40°C, xflow                    | 6.E-05        | 1/°C              | ASTM E 831  |
| HDT/Af, 1.8 MPa Flatw 80*10*4 sp=64mm        | 113           | °C                | ISO 75/Af   |
| <b>PHYSICAL</b>                              |               |                   |             |
| Density                                      | 1.29          | g/cm <sup>3</sup> | ASTM D 792  |
| Mold Shrinkage, flow, 24 hrs (5)             | 0.5 - 0.7     | %                 | ASTM D 955  |

1) Typical values only. Variations within normal tolerances are possible for various colours. All values are measured at least after 48 hours storage at 230C/50% relative humidity.  
All properties, except the melt volume rate are measured on injection moulded samples.  
All samples are prepared according to ISO 294.

2) Only typical data for material selection purpose. Not to be used for part or tool design.  
3) This rating is not intended to reflect hazards presented by this or any other material under actual fire conditions.  
4) Own measurement according to UL.  
5) Measurements made from laboratory test coupon. Actual shrinkage may vary outside of range due to differences in processing conditions, equipment, part geometry and tool design. It is recommended that mold shrinkage studies be performed with surrogate or legacy tooling prior to cutting tools for new molded article.  
6) Needs hard coat to consistently pass 60 sec Vertical Burn.

Source, GMD, Last Update: 01/12/2015

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| TYPICAL PROPERTIES <sup>1</sup>                       | TYPICAL VALUE | UNIT                    | STANDARD     |
|-------------------------------------------------------|---------------|-------------------------|--------------|
| <b>PHYSICAL</b>                                       |               |                         |              |
| Mold Shrinkage, xflow, 24 hrs (5)                     | 0.5 - 0.7     | %                       | ASTM D 955   |
| Moisture Absorption (23°C / 50% RH)                   | 0.05          | %                       | ISO 62       |
| Melt Volume Rate, MVR at 260°C/5.0 kg                 | 25            | cm <sup>3</sup> /10 min | ISO 1133     |
| Melt Volume Rate, MVR at 280°C/2.16 kg                | 27            | cm <sup>3</sup> /10 min | ISO 1133     |
| Melt Volume Rate, MVR at 280°C/21.6 kg                | 22            | cm <sup>3</sup> /10 min | ISO 1133     |
| <b>ELECTRICAL</b>                                     |               |                         |              |
| Volume Resistivity                                    | 1.2E+16       | Ohm-cm                  | ASTM D 257   |
| Surface Resistivity                                   | 1.2E+16       | Ohm                     | ASTM D 257   |
| Relative Permittivity, 1 GHz                          | 3.1           | -                       | ASTM D 150   |
| Dissipation Factor, 1 GHz                             | 0.006         | -                       | ASTM D 150   |
| Dielectric Constant, 1.1 GHz                          | 3.06          | -                       | SABIC Method |
| Dielectric Constant, 1.9 GHz                          | 3.08          | -                       | SABIC Method |
| Dielectric Constant, 5 GHz                            | 3.08          | -                       | SABIC Method |
| Dissipation Factor, 1.1 GHz                           | 0.0063        | -                       | SABIC Method |
| Dissipation Factor, 1.9 GHz                           | 0.0061        | -                       | SABIC Method |
| Dissipation Factor, 5 GHz                             | 0.0059        | -                       | SABIC Method |
| <b>FLAME CHARACTERISTICS</b>                          |               |                         |              |
| UL Recognized, 94V-2 Flame Class Rating (3)           | 1             | mm                      | UL 94        |
| <b>AFTER 40 CYCLES, SIMILAR TO USCAR-2, CLASS III</b> |               |                         |              |
| Tensile Strain, brk, Type I, 50 mm/min                | 70            | %                       | ASTM D 638   |
| <b>AFTER 40 CYCLES, SIMILAR TO USCAR-2, CLASS IV</b>  |               |                         |              |
| Tensile Stress, brk, Type I, 50 mm/min                | 44            | MPa                     | ASTM D 638   |

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| PROCESSING PARAMETERS       | TYPICAL VALUE | UNIT |
|-----------------------------|---------------|------|
| <b>Injection Molding</b>    |               |      |
| Drying Temperature          | 110 - 120     | °C   |
| Drying Time                 | 2 - 4         | hrs  |
| Maximum Moisture Content    | 0.02          | %    |
| Melt Temperature            | 260 - 280     | °C   |
| Nozzle Temperature          | 255 - 275     | °C   |
| Front - Zone 3 Temperature  | 260 - 280     | °C   |
| Middle - Zone 2 Temperature | 260 - 280     | °C   |
| Rear - Zone 1 Temperature   | 245 - 265     | °C   |
| Hopper Temperature          | 40 - 60       | °C   |
| Mold Temperature            | 80 - 140      | °C   |

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